

⚡ Lab 06: Cloud Run Functions Qwik Start - CLI

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Overview

Created, deployed, and tested a Cloud Run function using the command line instead of the console. The function is triggered by a Pub/Sub topic and logs a Hello message when a message is published to the topic.

Console vs CLI Comparison

Lab 05 - Console	Lab 06 - CLI
HTTPS trigger	Pub/Sub topic trigger
Inline code editor	index.js and package.json files
Cannot automate	Fully scriptable and automatable

Objectives

- Create a Cloud Run function using the CLI
- Deploy the function triggered by a Pub/Sub topic
- Test by publishing a message to the topic
- View function logs via CLI

Task 1 - Create the Function

Set the region, created a directory, wrote the function code in index.js, and set up package.json with dependencies.

```
gcloud config set run/region [REGION]
mkdir gcf_hello_world && cd $_
nano index.js
```

Function code (index.js):

```
const functions = require('@google-cloud/functions-framework');
functions.cloudEvent('helloPubSub', cloudEvent => {
  const base64name = cloudEvent.data.message.data;
  const name = base64name ? Buffer.from(base64name, 'base64').toString() : 'World';
  console.log(`Hello, ${name}!`);
});
```

```
npm install
```

terminal is available in cloud shell terminal: type `gcp-console` to open again [Dismiss](#)

```

student_02_6971cd94280f@cloudshell:~ (qwiklabs-gcp-02-73d801596d61)$ mkdir gcf_hello_world && cd $_
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$ nano index.js
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$ nano package.json
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$ npm install
npm warn EBADENGINE Unsupported engine {
npm warn EBADENGINE   package: 'cloudevents@8.0.3',
npm warn EBADENGINE   required: { node: '>=16 <=22' },
npm warn EBADENGINE   current: { node: 'v24.15.0', npm: '11.14.1' }
npm warn EBADENGINE }
npm warn deprecated uuid@8.3.2: uuid@10 and below is no longer supported. For ESM codebases, update to uuid@latest. For CommonJS codebases, use uuid@11 (but be aware this version will likely be deprecated in 2028).

added 144 packages, and audited 145 packages in 11s

37 packages are looking for funding
  run `npm fund` for details

3 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

```

```

npm notice To update run: npm install -g npm@11.15.0
npm notice
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$ npm audit fix --force
npm warn using --force Recommended protections disabled.
npm warn audit Updating @google-cloud/functions-framework to 2.0.0, which is a Ver major change.

removed 41 packages, changed 1 package, and audited 104 packages in 1s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$

```

Task 2 - Deploy the Function

```

gcloud functions deploy nodejs-pubsub-function --gen2 --runtime=nodejs20 --
region=[REGION] --source=. --entry-point=helloPubSub --trigger-topic cf-demo --
allow-unauthenticated
gcloud functions describe nodejs-pubsub-function --region=[REGION]

```

```

deployment-tool: cli-gcloud
name: projects/qwiklabs-gcp-02-73d801596d61/locations/us-central1/functions/nodejs-pubsub-function
satisfiesPzi: true
serviceConfig:
  allTrafficOnLatestRevision: true
  availableCpu: '0.1666'
  availableMemory: 256M
  environmentVariables:
    LOG_EXECUTION_ID: 'true'
  ingressSettings: ALLOW_ALL
  maxInstanceCount: 5
  maxInstanceRequestConcurrency: 1
  revision: nodejs-pubsub-function-00001-zac
  service: projects/qwiklabs-gcp-02-73d801596d61/locations/us-central1/services/nodejs-pubsub-function
  serviceAccountEmail: cloudfunctionsa@qwiklabs-gcp-02-73d801596d61.iam.gserviceaccount.com
  timeoutSeconds: 60
  uri: https://nodejs-pubsub-function-6rpmf6vl5a-uc.a.run.app
  state: ACTIVE
  updateTime: '2026-05-25T19:09:59.845736479Z'
  url: https://us-central1-qwiklabs-gcp-02-73d801596d61.cloudfunctions.net/nodejs-pubsub-function
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)
$ █

```

Task 3 - Test the Function

Published a message to the cf-demo Pub/Sub topic to trigger the function.

```
gcloud pubsub topics publish cf-demo --message="Cloud Function Gen2"
```

```

nodejs-function
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)
)$ gcloud pubsub topics publish cf-demo --message="Cloud Function Gen2"
messageIds:
- '19628694739738948'
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)
)$ █

```

Task 4 - View Logs

```
gcloud functions logs read nodejs-pubsub-function --region=[REGION]
```

```

- '19628694739738948'
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$ gcloud functions logs
pubsub-function \
--region=us-central1
messageIds:
- '19628694739738948'
student_02_6971cd94280f@cloudshell:~/gcf_hello_world (qwiklabs-gcp-02-73d801596d61)$ gcloud functions logs
pubsub-function \
--region=us-central1
LEVEL:
NAME: nodejs-pubsub-function
EXECUTION_ID:
TIME_UTC: 2026-05-25 19:10:53.373
LOG: Hello, Cloud Function Gen2!

LEVEL: I
NAME: nodejs-pubsub-function
EXECUTION_ID:
TIME_UTC: 2026-05-25 19:10:53.032
LOG:

LEVEL: I
NAME: nodejs-pubsub-function
EXECUTION_ID:
TIME_UTC: 2026-05-25 19:09:47.287
LOG: Default STARTUP TCP probe succeeded after 1 attempt for container "worker" on port 8080.

LEVEL: I
NAME: nodejs-pubsub-function
EXECUTION_ID:

```

Real World Application

- Automatically process uploaded files when they land in Cloud Storage
- Trigger data pipelines when new messages arrive in Pub/Sub
- Deploy functions via CI/CD pipelines without touching the console
- Event-driven microservices that scale automatically with demand

What I Learned

- Cloud Run functions can be triggered by Pub/Sub topics not just HTTP
- Functions are written as separate files and deployed via gcloud commands
- The --gen2 flag uses the second generation execution environment
- Publishing a message to the trigger topic automatically runs the function
- Logs may take up to 10 minutes to appear after function execution

Challenges & How I Solved Them

Found vulnerabilities after running npm install to install the package.json dependencies. The terminal warned about

security vulnerabilities in the installed packages. Fixed by running `npm audit fix --force` which automatically updated the vulnerable packages to safer versions and resolved all reported issues before deploying the function.

Resources

GCP Cloud Run Functions Docs: <https://cloud.google.com/functions/docs>
Lab ID: GSP080